SOIL INDEX PROPERTIES

Water content (w) = (Ww / Ws) 100%

Specific gravity (G) = (Ws / (Ww w))

Bulk density () = W / V

Dry density (d) = Ws / V

Saturated density (sat) = Wsat / V

Void ratio (e) = Vv / Vs

Porosity (n) = (Vv / V) 100%

Degree of saturation (S) = (Vw / Vv) 100%

Air content (ac) = (Va / Vv) 100%

CONSISTENCY LIMITS

Plasticity Index (Ip) = wL wP

Liquidity Index (IL) = (w wP) / Ip

Consistency Index (IC) = (wL w) / Ip

Flow Index (IF) = (w1 w2) / log(N2 / N1)

Toughness Index (IT) = Ip / IF

STRESS DISTRIBUTION

Vertical stress by point load: z = (3Qz) / [2(R + z)^(5/2)]

Newmark's influence chart: z = q Influence value

PERMEABILITY

Darcys Law: q = k i A

Coefficient of permeability:

- Constant head: k = (QL) / (A t h)
- Falling head: k = (2.3aL / At) log10(h1 / h2)

SEEPAGE ANALYSIS

Discharge (q) = k H / L B (for flow net)

Number of flow channels and equipotential drops

COMPRESSIBILITY & CONSOLIDATION

Compression index (Cc) = 0.009(wL 10)

Settlement (Sc) = (Cc H / (1 + e0)) log10(pf / pi)

Time factor (Tv) = cv t / H

SHEAR STRENGTH OF SOIL

Mohr-Coulomb: = c + tan

Unconfined compression: qu = P / A

Shear strength (undrained): su = qu / 2

EARTH PRESSURE

Rankine Active: Ka = (1 sin) / (1 + sin)

Rankine Passive: Kp = (1 + sin) / (1 sin)

At rest: K0 1 sin (for normally consolidated soil)

Lateral earth pressure: h = K v

BEARING CAPACITY

Terzaghis formula (general):

qult = cNc + qNq + 0.5BN

Net bearing capacity: qnet = qult Df

Safe bearing capacity: qsafe = qnet / FOS

SLOPE STABILITY

Factor of Safety (FOS) = Resisting Forces / Driving Forces

For = 0 soil: FOS = c L / W sin